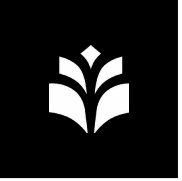


HTML

Web Programming I

Course materials originally by Sonsoles López-Pernas, adapted by Maria Kuismin



HTML

Student is able to
design and implement interactive and responsive web pages using HTML, CSS, JavaScript, and Bootstrap.

Content

Foundations of HTML

- What is HTML?
- How is an HTML document structured and what are the tags?

Structure and content elements

- How to add and emphasize text content?
- How to add images, videos, and other media to a page?

Navigation and user input and metadata

- How to link pages together and let users navigate?
- How to collect user input through forms?

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Foundations of HTML

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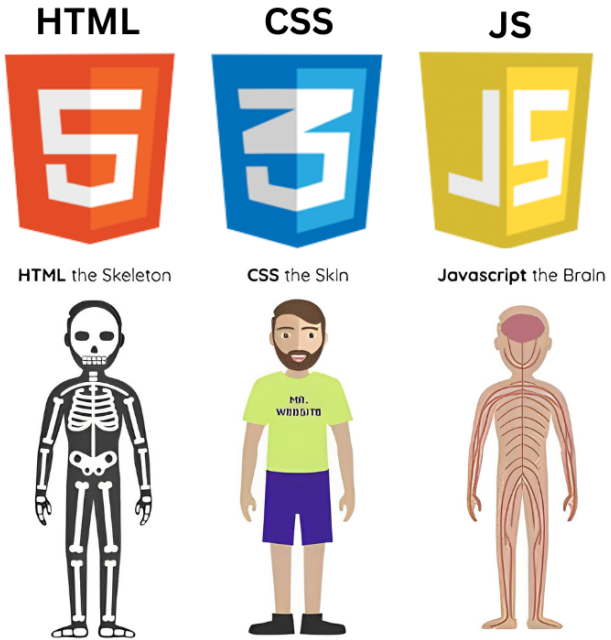
Structure and content elements

- How to add and emphasize text content?
- How to add images, videos, and other media to a page?

Navigation and user input and metadata

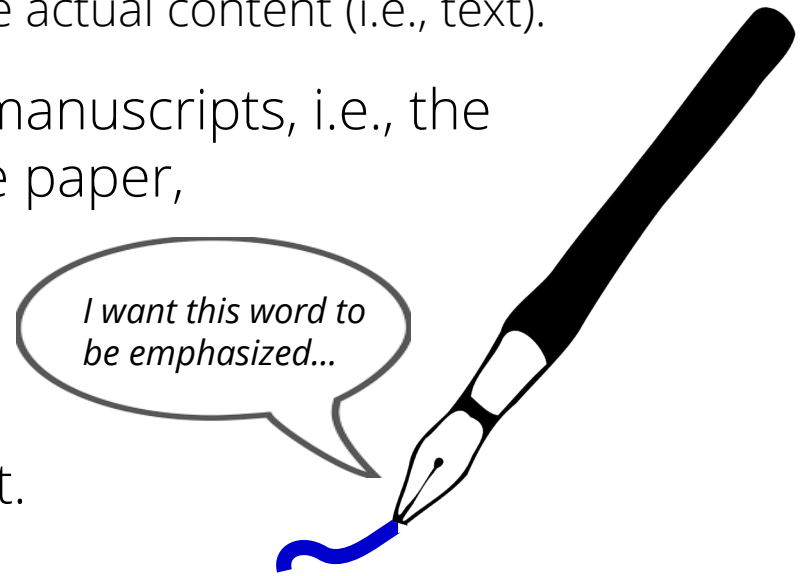
- How to link pages together and let users navigate?
- How to collect user input through forms?

Role of HTML in client technologies



HyperText Markup Language

- HTML stands for HyperText Markup Language.
  - Markup language = system for annotating a document in a way that is syntactically distinguishable from the actual content (i.e., text).
- Origin: "Marking up" of academic manuscripts, i.e., the instructions by editors to revise the paper, often written in blue on authors' manuscripts
- In HTML, markups are used for the semantic structuring of content.



Origin of HTML

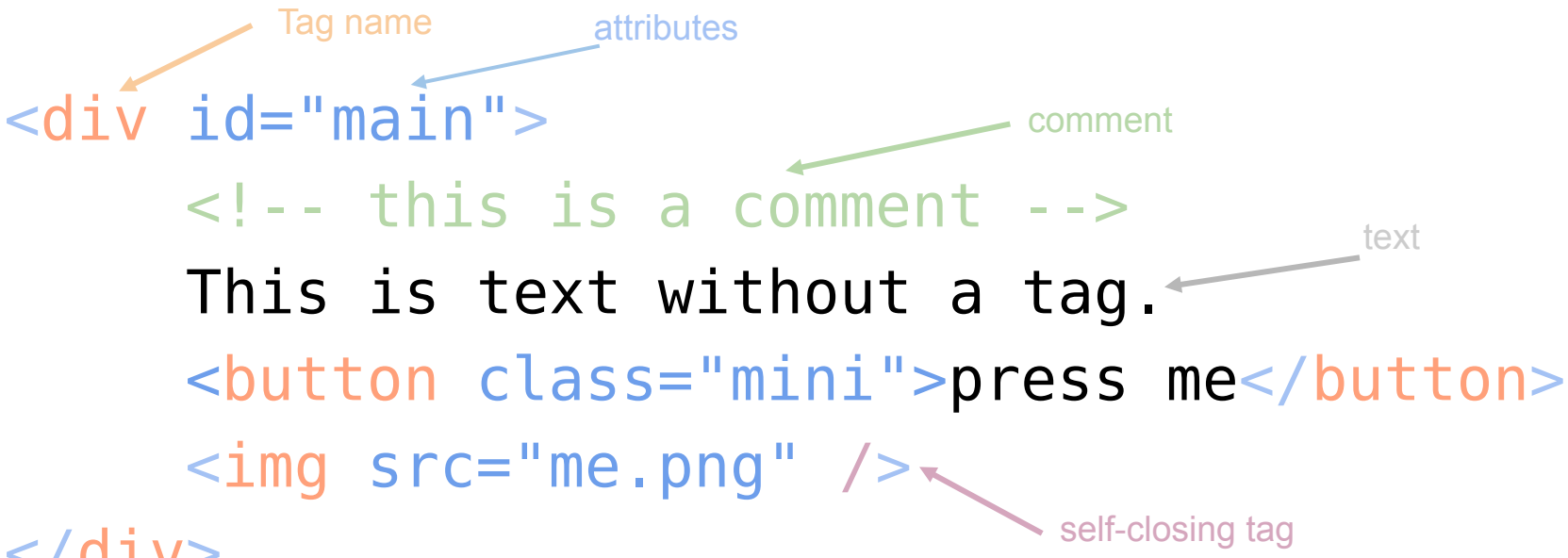
- In 1989 Tim Berners-Lee created the WWW (world wide web) project
- He defined HTML based on the SGML (standard generalized markup language) standard in 1993.
- The W3C develops technical specifications and guidelines, so that high technical and editorial quality can be ensured.



Documentation

- **MDN**: https://developer.mozilla.org/es/docs/HTML
- **W3Schools**: https://www.w3schools.com
- **W3C**: https://w3c.github.io/html-reference/
- **HTML5 Visual Cheat Sheet:** https://htmlcheatsheet.com/

HTML: syntax example



Structure of an HTML document

content type

document metadata

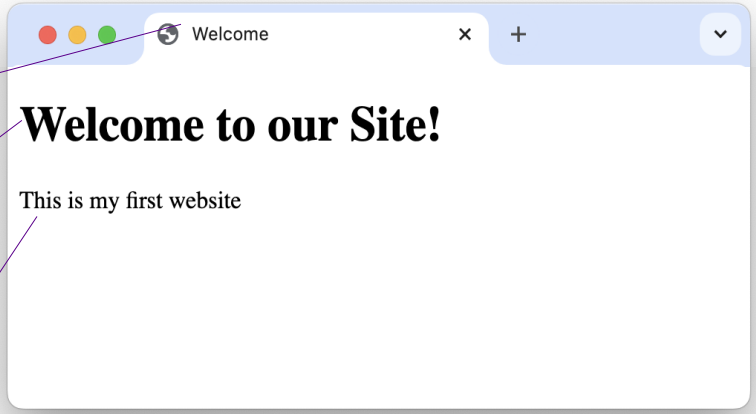
document content

```
<!doctype html>
<html>
  <head>
    <title> Welcome </title>
  </head>
  <body>
    <h1>Welcome to our Site! </h1>
    <p> This is my first website </p>
  </body>
</html>
```

document title

heading level 1

paragraph



HTML tags

basic tags, text formatting, headings, comments, links, images, layouts, forms, tables, lists, frames, audio, video & others

HTML Cheat Sheet

HTML Basic Tags

<!DOCTYPE> : Defines type of the document.
<html> ... </html> : Root of an HTML document.
<head> ... </head> : Container for all the metadata about an HTML document
<title> ... </title> : Defines title for the document.
<meta /> : Defines metadata like character set, viewport, keywords, page description, author etc...
<base /> : Specifies default URL for all links on a page.
<link /> : Defines link to external sources.
<style> ... </style> : Defines style for a document.
<script> ... </script> : Defines client-side scripts.
<noscript> ... </noscript> : Specifies an alternate content to be displayed if the browser doesn't support scripts.
<body> ... </body> : Represents the main body of a document.

HTML Tables

<table> ... </table> : Table
<tr> ... </tr> : Table Row
<th> ... </th> : Table Header Cell
<td> ... </td> : Table Data
<caption> ... </caption> : Table Caption
<thead> ... </thead> : Group of header contents
<tbody> ... </tbody> : Group of body contents
<tfoot> ... </tfoot> : Group of footer contents
<colgroup> ... </colgroup> : Group of columns
<col /> : Column in a <colgroup>

HTML Lists

 ... : Unordered List
 ... : List Item
 ... : Ordered List
<dl> ... </dl> : Description List
<dt> ... </dt> : Terms of the description list
<dd> ... </dd> : Describe each term of the description list

HTML Frames, Audio, Video & Others

<iframe> ... </iframe> : Inline Frame
<audio> ... </audio> : Audio
<video> ... </video> : Video
<source /> : Resources for <audio>, <video> and <picture> elements
<track /> : Text track for <audio> and <video> elements
<embed /> : Container for external applications
<object> ... </object> : Embedded Object
<param /> : Parameter for <object>

HTML Text Formatting

 ... : Paragraph

 : Single line break
<hr /> : Horizontal Rule
 ... : Bold Text
 ... : Italic Text
<small> ... </small> : Smaller Text
<mark> ... </mark> : Marked / Highlighted Text
<del> ... </del> : Deleted Text
<ins> ... </ins> : Inserted Text
 ... : Important Text
 ... : Emphasized Text
_{...} : Subscript
^{...} : Superscript
<blockquote> ... </blockquote> : Block Quotation
<q> ... </q> : Inline Quotation
<abbr> ... </abbr> : Abbreviations / Acronyms.
<address> ... </address> : Address
<cite> ... </cite> : Citation
<bdo> ... </bdo> : Changes text direction.
<code> ... </code> : Programming Code
<pre> ... </pre> : Preformatted Text
<samp> ... </samp> : Sample output of a program
<kbd> ... <kbd> : Keyboard Input
<var> ... </var> : Variable
<dfn> ... </dfn> : Definition
<meter> ... </meter> : Scalar Measurement
<progress> ... </progress> : Progress bar
<u> ... </u> : Underlined Text

HTML Forms

<form> ... </form> : HTML form
<input /> : Input Field
<label> ... </label> : Label for <input> element
<button> ... </button> : Clickable button
<textarea> ... </textarea> : Multiline Text Input
<select> ... </select> : Drop down list
<option> ... </option> : An option in drop down list
<optgroup> ... </optgroup> : Groups related options in dropdown list
<fieldset> ... </fieldset> : Groups related elements in a form
<legend> ... </legend> : Caption for <fieldset> element
<datalist> ... </datalist> : List of options
<output> ... </output> : Result

HTML Headings

<h1> ... </h1> : Heading One
<h2> ... </h2> : Heading Two
<h3> ... </h3> : Heading Three
<h4> ... </h4> : Heading Four
<h5> ... </h5> : Heading Five
<h6> ... </h6> : Heading Six

HTML Comments

<!-- Single Line comment -->
<!--
Multiple
Lines
comment
-->

HTML Links

<a> ... : Hyperlink

HTML Images

<img /> : Image
<map> ... </map> : Image Map with clickable area
<area /> : Clickable Area in <map>
<picture> ... </picture> : Group of pictures
<figure> ... </figure> : Figures / Diagrams / Illustrations
<figcaption> ... </figcaption> : Caption for <figure>
<canvas> ... </canvas> : Graphics
<svg> ... </svg> : SVG graphics

HTML Layouts

 ... : Division / Section / Block Container for HTML elements
 ... : Inline container for HTML elements
<header> ... </header> : Header content of a document
<main> ... </main> : Main content of a document
<footer> ... </footer> : Footer content of a document
<nav> ... </nav> : Navigation Links
<section> ... </section> : Section in a document
<article> ... </article> : Article in a document
<aside> ... </aside> : Side content in a document
<details> ... </details> : Additional details that user can hide or view on demand.
<summary> ... </summary> : Heading for <details> element
<dialog> ... </dialog> : Dialog Box

What do these tags do?

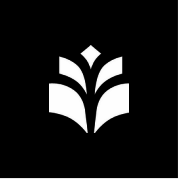
<button>Submit</button>

```
<ol>
    <li>First item</li>
    <li>Second item</li>
</ol>
```

UEF Website

** I am a paragraph **

<input type="text" placeholder="input stuff here">



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Text tags



- text tags for the semantic structuring of content
- texts are usually basically made up of
  - headlines <h1></h1>, <h2></h2>, … , <h6></h6>
    e.g., <h1>`Level 1 Headline`</h1>
  - paragraph `This is a paragraph example of how paragraphs must be formatted in an HTML document.`

Semantic vs. physical emphasis

Semantic emphasizers:

```
<em>emphasized text</em>
<strong> important text</strong>
<code> source code</code>
<dfn>definition</dfn>
<cite>title of a work</cite>
```

Physical emphasizers:

```
<small> smaller text </small>
<b>bold text</b>
<i>Italic text</i>
<u>underlined text</u>
<s>strikethrough text</s>
text<sub>subscript</sub>
text<sup>superscript</sup>
```

emphasized text
important text
`source code`
definition
title of a work
<small>smaller text</small>
bold text
Italic text
<u>underlined text</u>
~~strikethrough text~~
text$_{\text{subscript}}$
text^{superscript}

Other text

Preformatted paragraph (fixed-spaced font):

```
<pre>
for (i=1 ; i<10 ; i++) {
    print $i
}
</pre>
```

Textual quote

```
<blockquote cite="http://uef.fi">A university in Finland</blockquote>
```

Single line break

```
<br/>
```

HTML attributes

In addition to the tag name, HTML tags can have attributes. For example, for the image tag, we need to specify the source of the image:

```
<img src="cat.png"/>
```

Lists

Numbered list:

```
<ol>
   <li> Element 1 </li>
   <li> Element 2 </li>
   <li> Element 3 </li>
</ol>
```

1. Element 1
2. Element 2
3. Element 3

Unnumbered list:

```
<ul>
   <li> Element 1 </li>
   <li> Element 2 </li>
   <li> Element 3 </li>
</ul>
```

- Element 1
- Element 2
- Element 3

Tables

Tables are defined with the
<table> tag,
<tr> for each row,
<th> for header cell, and
<td> for each cell

	Population	Men	Women
Germany	82,020,578	40,346,853	41,673,725
France	65,578,819	31,764,615	33,814,204
United Kingdom	63,896,071	31,423,339	32,472,732

```
<table>
  <tr>
    <th> </th>
    <th> Population </th>
    <th> Men </th>
    <th> Women </th>
  </tr>
  <tr>
    <th> Germany </th>
    <td> 82,020,578 </td>
    <td> 40,346,853 </td>
    <td> 41,673,725 </td>
  </tr>
  <tr>
    <th> France </th>
    <td> 65,578,819 </td>
    <td> 31,764,615 </td>
    <td> 33,814,204 </td>
  </tr>
  <tr>
    <th> United Kingdom </th>
    <td> 63,896,071 </td>
    <td> 31,423,339 </td>
    <td> 32,472,732 </td>
  </tr>
</table>
```

Block and line elements

HTML elements can be displayed as block or inline by default.

Block: Block elements take up the entire width of the page and force a new line before and after, e.g.,<article> <aside> <canvas> <footer> <h1> <h2> <h3> <h4> <h5> <h6> <header> <p> <pre> <section> .
The element: A block element that is used to contain other HTML elements inside it. It has no meaning, it is semantically neutral.

Line: Inline elements only take up the width needed and do not force new lines, e.g. , <a>
 <img> <button> <input> <select> <textarea>...
The element: A line element that is used to contain other HTML elements inside it. It has no meaning, it is semantically neutral.

id and class attributes

id attribute
uniquely identifies an HTML element on a page.

An HTML element should only have one ID attribute and it must be unique on the page.

```
<div id="theBox">
<p id="content">Paragraph 1</p>
</div>
```

Class attribute
defines a class of HTML elements.

An element can have several CLASS attributes.

```
<div class="RedBackgroundBox ">
<p class="text">Paragraph 1</p>
</div>
```

Images, Video, Audio

Images used on the website: GIF, JPG, PNG, and vector images in SVG.

```
<img src="photo.jpg" width="470" height="600" alt="Self-portrait"/>
```

Responsive images uses picture tag.

Allows to use multiple images depending on the media conditions.

```
<picture>
<source media="(min-width: 1100px)" srcset="shaum6.jpg">
<source media="(min-width: 1000px)" srcset="shaum5.jpg">
<source media="(min-width: 800px)" srcset="shaum4.jpg">
<source media="(min-width: 600px)" srcset="shaum3.jpg">
<source media="(min-width: 400px)" srcset="shaum2.jpg">
<source media="(min-width: 300px)" srcset="shaum1.jpg">
<img src="shaum1.jpg" alt="shaum">
</picture>
```

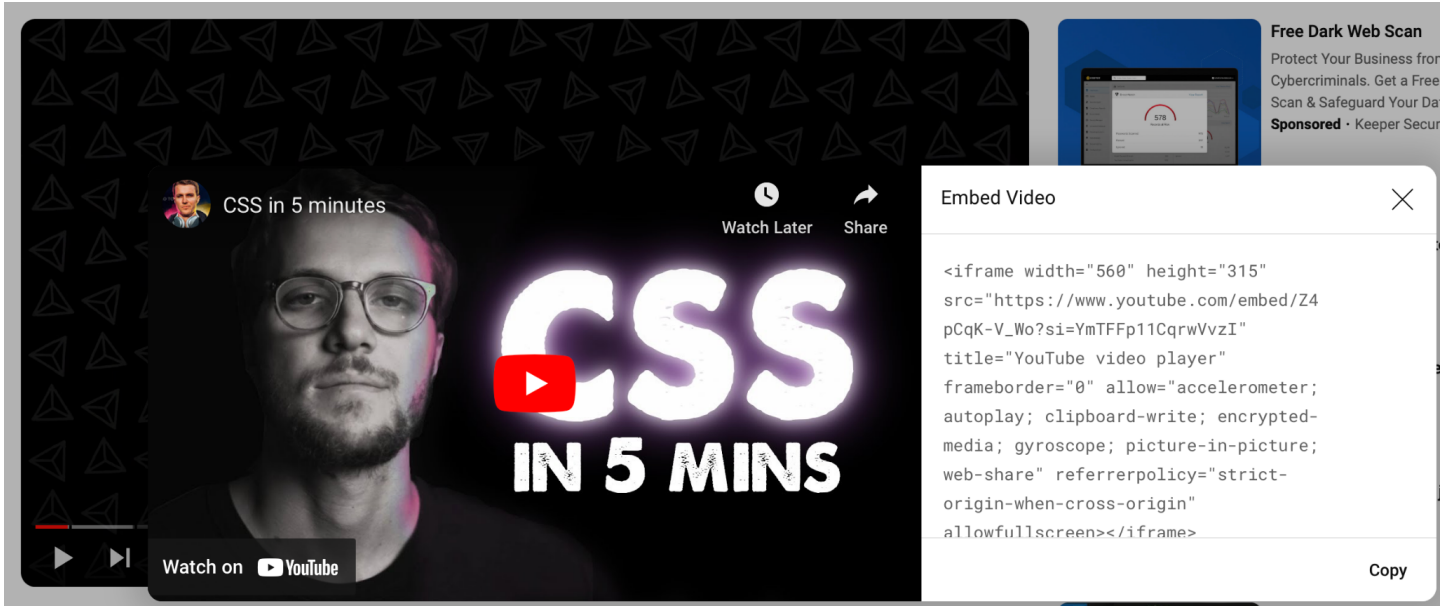


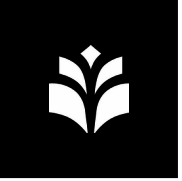
Video `<video src="video.mp4" poster="videojpg"autoplay loop controls></video>`

Audio `<audio src="audio.mp3" controls autoplay loop></audio>`

Embedding

```
<iframe width="560" height="315" allowfullscreen src="https://www.youtube.com/embed/Z4pCqK-V_Wo?si=lto94MJWIfzp-aDw"></iframe>
```





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HTML navigation

<a> tag is used to define links, whether local or external, to pages or resources.

- The main attribute is **href**, which contains the URL of the link destination.
- The target attribute allows us to decide where to open the link (e.g., **_blank** to open the link in a new window/tab)

Examples:

- Link to go to DuckDuckGo in a new tab.
  ```
  <a href="https://duckduckgo.com/" target="_blank">Go to DuckDuckGo</a>
  ```
- Sends a request to the email client for: name@myDomain.com
  ```
  <a href="mailto:name@myDomain.com">Contact</a>
  ```
- Link to open a PDF file stored in our server
  ```
  <a href="file2.pdf">File 2</a>
  ```

Menus

A menu is a list of links.

It is recommended to use the **<nav>** tag and the ** ** list tags and the **<a>** link.

```
<nav>
    <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="news.html">News</a></li>
        <li><a href="contact.html">Contact</a></li>
        <li><a href="about.html">About us</a></li>
    </ul>
</nav>
```

URLs

URL (Uniform Resource Locator) is the locator of a file on the Internet.

- **Absolute URLs:** absolute address of a page on the Internet.

```
<a href="https://news.ycombinator.com/">Hacker News</a>
```

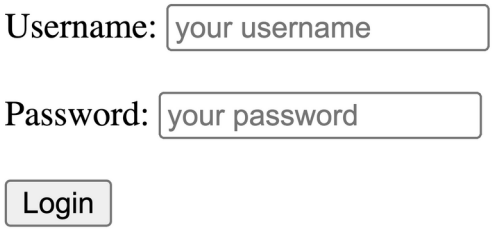
- **Local or relative URL:** relative path to a page or file relative to the server directory where the page is displayed.

```
<a href="pag2.html">See next</a>
<link href="css/style.css" rel="stylesheet" type="text/css"/>
<a href="pag/pag5.html">See example 5</a>
<a href="../index.html">Home page</a>
<img src="../imag/img1.png" alt="image"/>
```

Forms

Pages with user-fillable fields are created with the <form> tag. with more nested tags to create fields, selectors, buttons: <input>,<textarea>, <button>, <select>, <option>, <optgroup>, <fieldset>, <label> and <output>.

- allow to make HTTP requests by sending data to the web server
- action attribute of the form specifies the URL where the request must be sent.
- method attribute of the form specifies whether the request is GET or POST type.
  - In **GET** requests, data is sent in the URL query as a string of name=value pairs separated by &, escaping conflicting characters: *name* is the value of the name attribute of the form fields, and *value* is the content or what is indicated in the value attribute.
  - In **POST** requests, data is sent in the body of the request. It is encoded as specified in the form's enctype attribute: application/x-www-form-urlencoded, multipart/form-data , or text/plain.




```
<form action="https://elearn.uef.fi/login" method="post">

    <label for="username">Username:</label>
    <input type="text" id="username" name="username"
       value="" placeholder="your username"/>

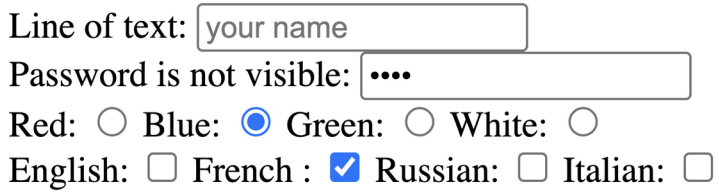
    <label for="passwd">Password:</label>
    <input type="password" id="passwd"
      name="password"  value=""
     placeholder="your password"/>

    <input type="submit" value="Login">

</form>
```

Forms: input types and validation

- <input> for text input, password, numbers, dates, upload files
- the type attribute indicates the type of field: color, date, email, file, number, password, radio, range, text (default)
- Attributes to validate the entered value
  - required to force a box to be filled. (required="true")
  - pattern to match the entered text with a regex (pattern=".+@.+")
  - min indicates the minimum value. (min="0")
  - max indicates the maximum value. (max="10")




```
<label for="name">Line of text:</label>
<input type="text" id="name" name="name" value="" placeholder="your name" />

<label for="password">Password is not visible:</label>
<input type="password" id="password" name="password" value="" placeholder="your password" />

<label for="red">Red:</label> <input type="radio" id="red" name="color" value="red">      ➞ exlusive choices
<label for="azul">Blue: </label> <input type="radio" for="blue" name="color" value="blue" checked>
<label for="green">Green:</label> <input type="radio" id ="green" name="color" value="green">
<label for="white">White:</label> <input type="radio" id="white" name="color" value="white ">

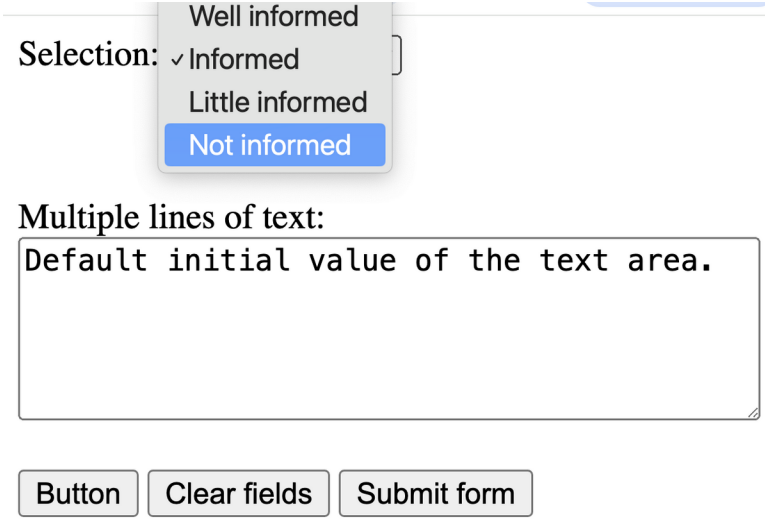
<label for="english">English:</label> <input type="checkbox" id="english" name="english" value="in">      ➞ non-exclusive choices
<label for="french">French :</label> <input type="checkbox" id="french" name="french" value="fr" checked>
<label for="russian">Russian:</label> <input type="checkbox" id="russian" name="russian" value="ru">
<label for="italian">Italian:</label> <input type="checkbox" id="italian" name="italian">
```

Form: other elements

<textarea> To enter a text with several lines.
<select> To create a drop-down list. Each option is described with the <option> tag
<button> Classic buttons, to send the form, and to clear the fields.

```
<label for="seleccion">Selection:</label>
<select id="seleccion" name="sel">          → dropdown menu
        <option>Well informed</option>
        <option selected>Informed</option>
        <option>Little informed</option>
        <option>Not informed</option>
</select>
<label for="area">Multiple lines of text:</label> <br />
<textarea id="area" name="msg" rows="5" cols="40">   → long text
        Default initial value of the text area.
</textarea>

<button type="button">Button </button>  → buttons
<button type="reset">Clear fields</button>
<button type="submit">Submit form</button>
```

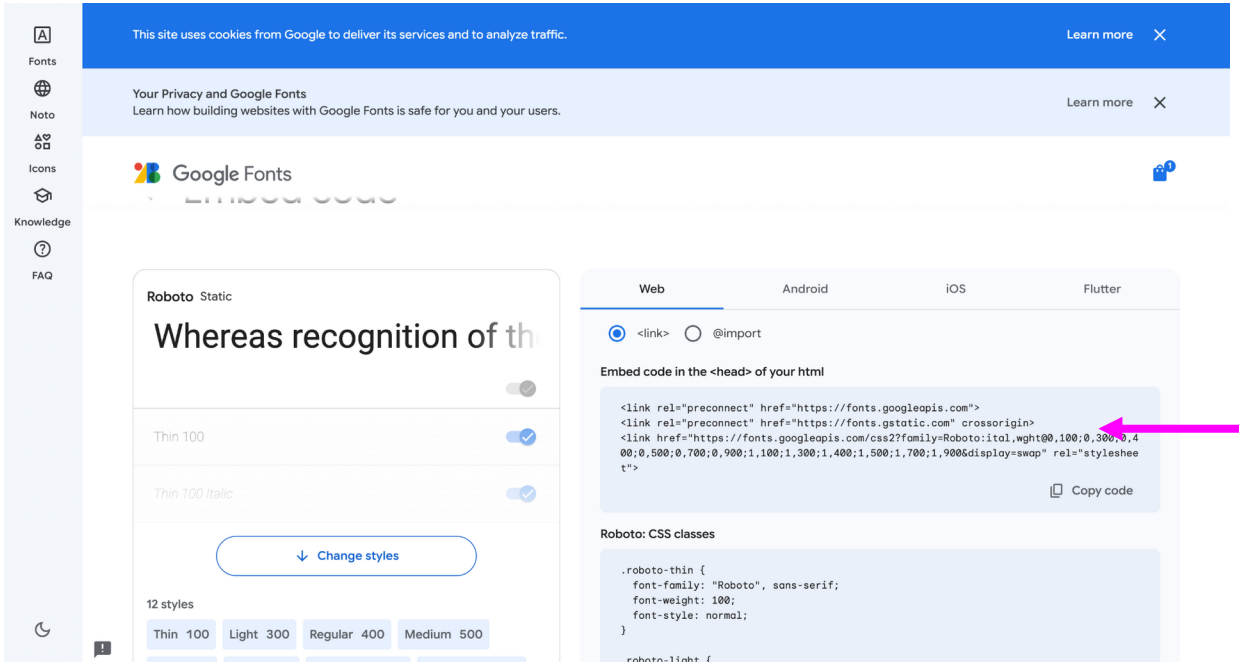


HTML head: metadata

Encoding: `<meta charset="UTF-8">`
Description: `<meta name="description" content="Free Web tutorials">`
Keywords: `<meta name="keywords" content="HTML, CSS, JavaScript">`
Author: `<meta name="author" content="John Doe">`
Responsive: `<meta name="viewport" content="width=device-width, initial-scale=1.0">`

og - Open Graph
Title (when link sharing): `<meta property="og:title" content="Itä-Suomen yliopisto">`
Description (when link sharing): `<meta property="og:description" content="Welcome to UEF.">`
Preview image (when link sharing): `<meta property="og:image" content="...">`
Preview image (when link on X): `<meta name="twitter:description" content="Welcome to UEF.">`
Preview image (when link on X): `<meta name="twitter:title" content="Itä-Suomen yliopisto">`

Adding custom fonts



Adding this code to the **<head/>** of the HTML file

Example: Cat page

Welcome to Kažigram

- Feed
- New post
- About



The winter is coming

Still some grass to eat.

♡ Like

Share your cat moment

Title:

Caption:

Photo: Selaa... Ei valittua tiedostoa.

Post

About Kažigram

Kažigram is a place to share cat moments.

© 2026 Kažigram

```
<!DOCTYPE html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <meta name="description"
content="Kažigram — the cat-themed page">
  <title>Kažigram</title>
</head>

<body>

    ------------------->

</body>


</html>
```

```
<body>
  <header>
    <h1>Welcome to Kažigram</h1>
    <nav>
      <ul>
        <li><a href="#feed">Feed</a></li>
        <li><a href="#new">New post</a></li>
        <li><a href="#about">About</a></li>
      </ul>
    </nav>
  </header>

  <section id="feed">
    <article>
      <img src="cat1.jpg" alt="Cat outside, some snow and grass">
      <h2>The winter is coming</h2>
      <p>Still some grass to eat.</p>
      <button>♡ Like</button>
    </article>

  </section>
  <!-- The form does not yet have any functionality. -->
  <section id="new">
    <h2>Share your cat moment</h2>
    <form action="#" method="post">
      <div>
        <label for="title">Title:</label>
        <input type="text" id="title" name="title" required>
      </div>
      <div>
        <label for="caption">Caption:</label>
        <textarea id="caption" name="caption" rows="3"
required></textarea>
      </div>
      <div>
        <label for="photo">Photo:</label>
        <input type="file" id="photo" name="photo" accept="image/*"
required>
      </div>
      <button type="submit">Post</button>
    </form>
  </section>

  <footer>
    <div id="about">
      <h3>About Kažigram</h3>
      <p>Kažigram is a place to share cat moments.</p>
    </div>
    <p>&copy; 2026 Kažigram</p>
  </footer>
</body>
```

Quick check

Go to

www.menti.com

Enter the code

2293 3778



Or use QR code

Summary

- HTML (HyperText Markup Language) it's used for the semantic structuring of content
- HTML document contains head (metadata) and body (visible content); tags can have attributes, content, and may be self-closing
- text content is added with semantic tags
- images, video, and audio are added with <img>, <video>, and <audio>; id (unique) and class (reusable) attributes connect HTML to CSS and JavaScript
- pages are linked with <a> tags; menus combine <nav>, <ul>, <li>, and <a>
- forms collect user input through <form> with <input>, <textarea>, <select>, and <button> — and the action and method attributes determine where and how data is sent
- head section contains metadata like character encoding, and Open Graph tags for social media sharing

Bibliography

Materials of the course *Computación en Red*
- Universidad Politécnica de Madrid

https://developer.mozilla.org/en-US/docs/Web/HTML

https://www.w3schools.com/html/